Depropanizer Column Ver 1 · OTS

DC3-V1-ME-EQL-001

Equipment list

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-ME-EQL-001
Title	Equipment list
Discipline	Mechanical
Originator / DRAWN	Apinan Soottitantawat
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Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

1 · Purpose

Equipment list of the C-201 depropanizer unit. Sizes come from the simulator model (volumes and UA values) and the Design tab sizing; operating conditions are the highest of the three cases of the heat and material balance DC3-V1-PE-HMB-001; design conditions follow the design-basis rules below. Thicknesses and weights are FEED estimates for the vendor to confirm.

2 · Equipment list

Tag	Name	Type	Qty	Size	Side	Op. P / T (design)	Max op. P / T	Design P MPa g / T °C	MDMT °C	Material	Insulation	Shell / head mm	Empty kg	Notes
C-201	Depropanizer column	Trayed column, 27 sieve trays (tray type assumed), 2:1 ellipsoidal heads, skirt-supported	1	1.7 m ID × 23.0 m T/T; tray spacing 0.61 m; feed tray 18	Shell — top	1.700 / 45.0	1.702 / 45.1	**1.90 / 140**	−46	SA-516 Gr. 70 (impact tested); trays CS (assumed)	H 50 mm mineral wool (heat conservation), fire-safe on the skirt (assumed)	18 / 18	23,200	Diameter and height from the Design tab sizing; design pressure rule (1) at the top, rule (4) at the bottom; PSV-201 on the overhead line set 1.80 MPa g.
					Shell — bottom	1.718 / 109.0	1.722 / 109.1	**1.95 / 140**	−46					
V-201	Feed surge drum	Horizontal drum, 2:1 ellipsoidal heads	1	2.1 m ID × 6.5 m T/T (24.8 m³ from the model, L/D ≈ 3 assumed)	Shell	0.400 / −1.0	0.400 / −1.0	**1.35 / 65**	−46	SA-516 Gr. 70 (impact tested)	AC 40 mm anti-condensation (operates at −1 °C, assumed)	16 / 16	7,400	Design pressure set by rule (2): feed bubble pressure at 50 °C 1.30 MPa g. No PSV shown on the P&ID (note 11, HOLD).
V-202	Reflux drum	Horizontal drum, 2:1 ellipsoidal heads	1	1.8 m ID × 5.6 m T/T (15.7 m³ from the model)	Shell	1.680 / 39.4	1.691 / 41.8	**1.90 / 75**	−46	SA-516 Gr. 70 (impact tested)	None (personnel protection not required below 60 °C)	20 / 18	6,700	Protected by PSV-201 through E-202 (no isolation valve between C-201 and V-202).
E-201	Feed preheater	Shell & tube, TEMA BEU (feed in the U-tubes, LP steam in the shell) — assumed	1	90 m² (= model UA 45 kW/K ÷ U 500 W/m² K assumed); duty 3.50 MW	Shell (LP steam)	0.148 / 127.5	0.260 / 140.2	**0.70 / 175**	0	Shell SA-516 Gr. 70; tubes SA-179 seamless CS (assumed)	H 50 mm (assumed)	—	—	Tube side on the P-201 discharge: design pressure rule (3) (shut-off 2.72 MPa g).
					Tubes (feed)	2.363 / 75.0	2.563 / 75.0	**2.85 / 175**	−46					
E-202	Overhead condenser	Shell & tube, TEMA AES (condensing propane in the shell, cooling water in the tubes) — assumed	1	290 m² (= model clean UA 203 kW/K ÷ U 700 W/m² K assumed); duty 3.96 MW	Shell (overhead)	1.700 / 45.0	1.702 / 45.1	**1.90 / 75**	−46	Shell SA-516 Gr. 70; tubes SA-179 CS, CW side CA 3 mm (assumed)	None	—	—	CW side design pressure 0.70 MPa g is below 10/13 of the shell side — tube-rupture case for the Relief tab.
					Tubes (cooling water)	0.450 / 25.3	0.450 / 26.8	**0.70 / 75**	0					
E-203	Reboiler	Vertical thermosyphon, TEMA BEM (process in the tubes, LP steam in the shell) — assumed	1	73 m² (= model UA 73 kW/K ÷ U 1000 W/m² K assumed); duty 2.24 MW	Shell (LP steam)	0.254 / 139.7	0.304 / 144.3	**0.70 / 175**	0	Shell SA-516 Gr. 70; tubes SA-179 seamless CS (assumed)	H 50 mm (assumed)	—	—	Circulation rate is not modelled; 30 % mass vaporized at the outlet assumed for the line sizing.
					Tubes (C-201 bottoms)	1.736 / 109.0	1.740 / 109.1	**1.95 / 175**	−46					
P-201A/B	Feed pumps	Centrifugal, API 610 between-bearings multistage (head ≈ 361 m) — type assumed; A motor, B steam turbine	2	Rated 77 m³/h × 1.71 MPa (317 m); absorbed 56.2 kW at rated; motor 75 kW	Casing	2.363 / −1.0	2.563 / −1.0	**2.85 / 65**	−46	CS casing, impact tested; 12 % Cr internals (assumed)	None	—	—	Pump curve and system from the model (H0 2300 kPa, a 0.1 kPa/(m³/h)²). P-201B: steam turbine drive (diverse drive, P&ID note 7).
P-202A/B	Reflux / product pumps	Centrifugal, API 610 OH2 single stage — type assumed; motor drives	2	Rated 102.5 m³/h × 0.462 MPa (100 m); absorbed 22 kW at rated; motor 30 kW	Casing	2.262 / 39.4	2.355 / 41.8	**2.60 / 70**	−46	CS casing, impact tested; 12 % Cr internals (assumed)	None	—	—	Pump curve from the model (H0R 700 kPa, a 0.0226). PDSL-230 auto-starts P-202B.

3 · Design basis applied

Design pressure rules

- (1) Vessels and exchangers: design pressure = max(1.10 × maximum operating pressure, maximum operating pressure + 0.17 MPa), rounded up to 0.05 MPa; maximum operating pressure = the highest value over the three HMB cases.
- (2) Vessels holding LPG: not below the bubble (vapour) pressure of the contents at the maximum metal temperature 50 °C (API 2510 practice, assumed).
- (3) Pump discharge systems: not below the shut-off pressure = maximum operating suction pressure + shut-off differential of the pump curve.
- (4) Lower sections of the column: design pressure at the top + column ΔP (PDI-212 high alarm) + static head of liquid to the reboiler return nozzle.
- (5) Utility sides: LP steam 0.70 MPa g, cooling water 0.70 MPa g (assumed system design pressures).
- (6) The PSV set pressure must not exceed the design pressure (MAWP) of the protected equipment (ASME VIII UG-134).

Design temperature rules

- (a) Design temperature = maximum operating temperature + 28 °C, rounded up to 5 °C.
- (b) Both sides of a steam-heated exchanger: saturation temperature at the steam design pressure (0.70 MPa g → 170 °C), rounded up to 175 °C.
- (c) Uninsulated vessels holding LPG: not below 65 °C (solar gain, assumed).
- (d) Minimum design metal temperature (MDMT) −46 °C for all LPG-containing equipment and piping (auto-refrigeration of propane on depressuring: normal boiling point −42 °C); 0 °C for utility sides (assumed).

Materials and thickness basis

- SA-516 Gr. 70 plate, allowable stress 138 MPa up to 343 °C (ASME II-D), joint efficiency 0.85 (spot radiography), corrosion allowance 3 mm, mill under-tolerance 0.3 mm (all assumed); impact tested to −46 °C (LPG service).
- Shell thickness ASME VIII Div. 1 UG-27(c)(1) (hoop stress governs), 2:1 ellipsoidal heads UG-32(d), minimum UG-16(b); nominal = next standard plate after the mill under-tolerance; MAWP back-calculated in the corroded condition; hydrotest UG-99(b) 1.3 × design pressure (stress ratio 1).
- Vessel sizes from the model volumes with L/D ≈ 3 and 2:1 heads; exchanger areas = model UA ÷ the Design tab's service U (500 / 700 / 1000 W/m² K for E-201 / E-202 / E-203, assumed).
- External pressure (vacuum), wind, seismic, nozzle loads and local stresses are not checked (query EM-06).

Results of the rules

Item	Rule	Result
C-201 top / V-202 / E-202 shell	(1) max(1.10 × 1.702, + 0.17 MPa)	1.90 MPa g
C-201 bottom / E-203 tubes	(4) top + 30 kPa ΔP + 18.1 kPa liquid head	1.948 / 1.95 MPa g
V-201	(2) feed bubble pressure at 50 °C 1.30 MPa g (rule 1: 0.60)	1.35 MPa g
P-201 casing, E-201 tubes, feed lines	(3) shut-off 2.723 MPa g; 1.10 × max operating 2.563	2.85 MPa g
P-202 casing, reflux / product lines	(3) shut-off 2.421 MPa g; 1.10 × max operating 2.355	2.60 MPa g
Steam-heated sides	(b) T_sat at 0.70 MPa g = 170.7 °C	175 °C
C-201 / V-202 / V-201	(a) max operating + 28 °C; (c) 65 °C minimum for uninsulated LPG	140 / 75 / 65 °C

Engineering queries raised

No.	Subject	Query	Proposed answer	Status
EM-01	Minimum design metal temperature and material	Depressuring LPG auto-refrigerates to about −42 °C; MDMT −46 °C was applied to C-201, V-201, V-202, E-202 shell, the pumps and the hydrocarbon piping. Plain SA-516-70 is not exempt from impact testing at this MDMT (UCS-66).	Specify SA-516-70N impact tested at −46 °C (and LT piping classes A1L / B1L); confirm the depressuring end temperature in the Relief tab.	OPEN
EM-02	V-201 design pressure and overpressure protection	Rule (2) sets V-201 at 1.35 MPa g (feed bubble pressure 1.30 MPa g at 50 °C) while it operates at 0.40 MPa g; the P&ID shows no PSV on V-201 (note 11).	Add a PSV on V-201 set at 1.35 MPa g (fire case and upstream blocked-in); confirm the upstream maximum pressure.	OPEN
EM-03	E-202 tube rupture (cooling-water side)	The CW side of E-202 is designed for 0.70 MPa g, below 10/13 of the shell design pressure (1.46 MPa g): a tube rupture would overpressure the cooling-water system with propane.	Raise the CW side to 1.50 MPa g or provide tube-rupture relief (TSV-216 recommended in the Safety tab) — Relief tab to size.	OPEN
EM-04	E-201 / E-203 steam side vs process side	The LP-steam shells (0.70 MPa g) are below 10/13 of the tube-side design pressure (E-201 2.85, E-203 1.95 MPa g): a tube rupture sends hydrocarbon into the steam / condensate system.	Tube-rupture case for both shells in the Relief tab; hydrocarbon detection on the condensate return (recommended addition).	OPEN
EM-05	Pump discharge design pressure basis	P-201 discharge items (E-201 tubes, 4"-P-202, 6"-P-203) use the shut-off pressure at the normal suction pressure (2.72 MPa g). If V-201 can reach its design pressure 1.35 MPa g, the shut-off rises to 3.67 MPa g.	Decide whether the maximum suction pressure is the V-201 operating or design pressure; with the latter the discharge piping moves to class 300 at 3.70 MPa g.	OPEN
EM-06	External pressure, wind and seismic not checked	C-201 (1.7 m × 23.0 m) is checked for internal pressure only. Steam-out and cooling of a blocked-in column can pull vacuum; wind and seismic loads set the bottom-course and skirt thickness of tall columns.	Design C-201, V-201 and V-202 for full vacuum (UG-28) or provide vacuum protection; civil/structural to run wind and seismic checks of C-201 and its skirt.	OPEN